

Design and Analysis of a 3D-Printed Planetary Gear Mechanism

ME 470 - Mechanical Design and Manufacturing II

Honors Option Project

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1 Introduction

This project focuses on the design, analysis, and fabrication of a 3D-printed planetary gear mechanism. Planetary gear systems are widely used in mechanical applications due to their ability to provide high torque transmission and compact speed reduction within a single stage.

The objective of this project was to develop a functional planetary gear system that demonstrates the fundamental relationships between gear speed and torque. This was accomplished through analytical derivation, computational modeling in Python, CAD design, and physical fabrication using additive manufacturing. A motion study was also performed to validate the predicted behavior of the system.

The final design was selected based on its ability to clearly demonstrate gear reduction and torque amplification while satisfying geometric constraints required for proper meshing.

2 Analysis

2.1 Purpose of Analysis

The purpose of this analysis is to derive the governing speed and torque relationships for the planetary gear mechanism. These equations are used to define the kinematic behavior of the system and will later support the computational model, CAD design validation, and final testing.

2.2 Planetary Gear Definitions

The simple planetary gear set considered in this project consists of a sun gear, planet gears, ring gear, and carrier. The following variables are used throughout the derivation.

ω_s = angular velocity of the sun gear

ω_p = angular velocity of the planet gear

ω_r = angular velocity of the ring gear

ω_c = angular velocity of the carrier

N_s = number of teeth on the sun gear

N_p = number of teeth on the planet gear

N_r = number of teeth on the ring gear

T_s = torque on the sun gear

T_r = torque on the ring gear

T_c = torque on the carrier

2.3 Speed Relationship Derivation

2.3.1 Relative Motion Approach

Because the planet gears both rotate about their own axes and revolve with the carrier, planetary gear systems are analyzed using relative angular velocities with respect to the carrier. This allows the gear train to be treated similarly to a conventional fixed-axis gear train.

2.3.2 Sun–planet mesh

The sun and planet gears form an external gear mesh, so their relative angular velocity ratio is

$$\frac{\omega_s - \omega_c}{\omega_p - \omega_c} = -\frac{N_p}{N_s}$$

Rearranging,

$$\omega_p - \omega_c = -\frac{N_s}{N_p}(\omega_s - \omega_c)$$

2.3.3 Planet–ring mesh

The planet and ring gears form an internal gear mesh, so their relative angular velocity ratio is

$$\frac{\omega_r - \omega_c}{\omega_p - \omega_c} = \frac{N_p}{N_r}$$

Rearranging,

$$\omega_r - \omega_c = \frac{N_p}{N_r}(\omega_p - \omega_c)$$

2.3.4 Overall Speed Equation

Substituting the sun–planet relation into the planet–ring relation gives

$$\omega_r - \omega_c = \frac{N_p}{N_r} \left[-\frac{N_s}{N_p}(\omega_s - \omega_c) \right]$$

This simplifies to

$$\omega_r - \omega_c = -\frac{N_s}{N_r}(\omega_s - \omega_c)$$

Therefore, the main speed relationship for the planetary gear set is

$$\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = -\frac{N_r}{N_s}$$

This is the primary speed equation used for the planetary gear mechanism.

2.3.5 Useful Rearranged Forms

For practical analysis, the main speed equation can be rearranged depending on which member is treated as the input, output, or fixed member.

Solve for ring speed

$$\omega_r = \omega_c - \frac{N_s}{N_r}(\omega_s - \omega_c)$$

Equivalent expanded form:

$$\omega_r = \omega_c \left(1 + \frac{N_s}{N_r}\right) - \frac{N_s}{N_r} \omega_s$$

Solve for sun speed

$$\omega_s = \omega_c - \frac{N_r}{N_s}(\omega_r - \omega_c)$$

2.4 Geometric Constraint

For proper meshing of the sun, planet, and ring gears, the tooth counts must satisfy the standard planetary gear geometry condition below.

$$N_r = N_s + 2N_p$$

This ensures that the planet gears can simultaneously mesh with both the sun and ring gears.

2.5 Torque Relationship Derivation

2.5.1 Power Balance

Assuming ideal operation with negligible losses, the planetary gear train must satisfy conservation of power. Therefore, the total power entering and leaving the system must balance.

$$T_s \omega_s + T_r \omega_r + T_c \omega_c = 0$$

2.5.2 Torque Ratio

Using the carrier as the moving reference frame, the power relation between the sun and ring can be written in terms of relative angular velocity.

$$T_s(\omega_s - \omega_c) + T_r(\omega_r - \omega_c) = 0$$

Substitute the speed relation

$$\omega_r - \omega_c = -\frac{N_s}{N_r}(\omega_s - \omega_c)$$

This gives

$$T_s(\omega_s - \omega_c) + T_r \left[-\frac{N_s}{N_r}(\omega_s - \omega_c) \right] = 0$$

Factoring out the common term yields

$$(\omega_s - \omega_c) \left[T_s - T_r \frac{N_s}{N_r} \right] = 0$$

Therefore,

$$T_s - T_r \frac{N_s}{N_r} = 0$$

and

$$\frac{T_r}{T_s} = \frac{N_r}{N_s}$$

So the ideal torque relationship is

$$T_r = \frac{N_r}{N_s} T_s$$

This shows that torque is scaled according to the gear tooth ratio.

2.5.3 Carrier Torque Equilibrium

For steady operation, the total external torque on the planetary gear train must also balance.

$$T_s + T_r + T_c = 0$$

So the carrier torque is

$$T_c = -(T_s + T_r)$$

Substituting the torque ratio gives

$$T_c = -\left(T_s + \frac{N_r}{N_s}T_s\right)$$

which simplifies to

$$T_c = -T_s\left(1 + \frac{N_r}{N_s}\right)$$

This expression is useful for evaluating carrier loading and output torque behavior.

2.6 Final Governing Equations

The final governing equations used in the analysis of the planetary gear mechanism are summarized below.

$$\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = -\frac{N_r}{N_s}$$

$$N_r = N_s + 2N_p$$

$$T_s\omega_s + T_r\omega_r + T_c\omega_c = 0$$

$$\frac{T_r}{T_s} = \frac{N_r}{N_s}$$

$$T_c = -T_s\left(1 + \frac{N_r}{N_s}\right)$$

These equations form the analytical basis for the computational model developed in Python.

3 Computational Model Development (Python)

3.1 Purpose of the Computational Model

A Python-based computational model was developed to evaluate the kinematic and torque behavior of the planetary gear mechanism before CAD modeling and fabrication. The model was used to verify gear geometry, calculate output speeds for multiple operating modes, and compare candidate gear sets for final design selection.

3.2 Operating Cases Evaluated

3.2.1 Case 1: Ring Fixed

- Fixed member: Ring gear
- Input: Sun gear
- Output: Carrier

3.2.2 Case 2: Sun Fixed

- Fixed member: Sun gear
- Input: Carrier
- Output: Ring

3.2.3 Case 3: Carrier Fixed

- Fixed member: Carrier
- Input: Sun gear
- Output: Ring

3.3 Candidate Gear Sets Evaluated

Three candidate gear sets were tested in the computational model to compare speed ratio behavior and support final design selection.

Table 2.1: Three candidate gear sets

Design	N_s	N_p	N_r	N_r/N_s	Notes
1	20	20	60	3.00	Baseline
2	15	20	55	3.67	Higher reduction
3	30	15	60	2.00	Faster output

3.4 Computational Results

Table 2.2 summarizes the predicted output speeds and torque behavior for each candidate gear set under the three standard operating cases. Figures 2.1 – 2.6 show the gear speed and torque relationships over a wide range of inputs.

Table 2.2: Results for different cases and inputs

Design	Case	Input	Output	Result
1	Ring Fixed	$\omega_s = 100$ rpm cw	ω_c	25.00 rpm cw
1	Sun Fixed	$\omega_c = 100$ rpm cw	ω_r	133.33 rpm cw
1	Carrier Fixed	$\omega_s = 100$ rpm cw	ω_r	33.33 rpm ccw
1	Torque Example	$\tau_s = 10$ Nm	τ_r, τ_c	30.00, -40.00 Nm
2	Ring Fixed	$\omega_s = 100$ rpm cw	ω_c	21.43 rpm cw
2	Sun Fixed	$\omega_c = 100$ rpm cw	ω_r	127.27 rpm cw
2	Carrier Fixed	$\omega_s = 100$ rpm cw	ω_r	27.27 rpm ccw
2	Torque Example	$\tau_s = 10$ Nm	τ_r, τ_c	36.67, -46.67 Nm
3	Ring Fixed	$\omega_s = 100$ rpm cw	ω_c	33.33 rpm cw
3	Sun Fixed	$\omega_c = 100$ rpm cw	ω_r	150.00 rpm cw
3	Carrier Fixed	$\omega_s = 100$ rpm cw	ω_r	50.00 rpm ccw
3	Torque Example	$\tau_s = 10$ Nm	τ_r, τ_c	20.00, -30.00 Nm

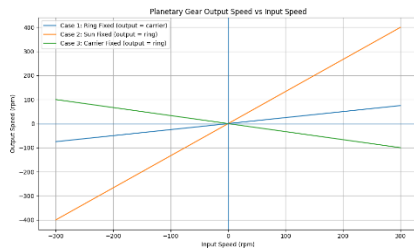


Figure 3.1: Case 1 gear output/input speed

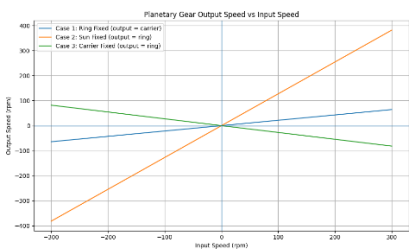


Figure 3.2: Case 2 gear output/input speed

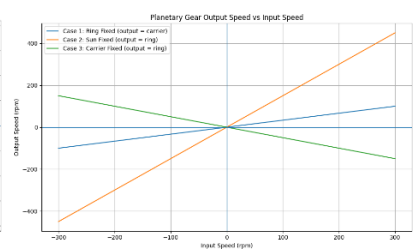


Figure 3.3: Case 3 gear output/input speed

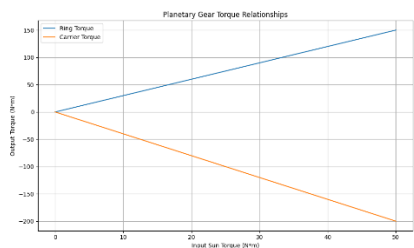


Figure 3.4: Case 1 torque relationship

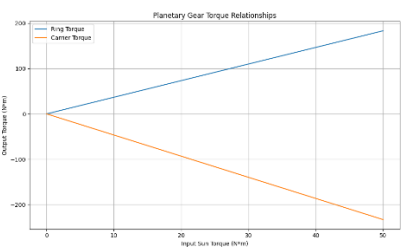


Figure 3.5: Case 2 torque relationship

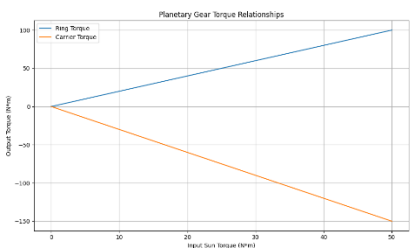


Figure 3.6: Case 3 torque relationship

3.5 Final Gear Selection

The final gear set selected for this project was $N_s = 30$, $N_p = 15$, and $N_r = 60$. This design was chosen because it produced the highest reduction ratio and the greatest ideal torque multiplication of the candidate gear sets tested. As a result, it provides the clearest

demonstration of planetary gear reduction behavior while still satisfying the required geometric constraints for proper meshing.

3.6 Python Implementation

The governing equations derived in the analysis section were implemented in Python to automate speed and torque calculations for the planetary gear mechanism. The script was used to evaluate multiple operating cases and compare candidate gear sets prior to CAD modeling and fabrication. The full source code is provided in appendix A.

4 Motion Study (SolidWorks)

A motion study was performed in SolidWorks to validate the analytical and computational results. Gear mates were applied to represent meshing between components, and a rotary motor was used to drive the input member. The resulting angular velocities were measured and compared to the predicted values from the computational model. The simulation results showed good agreement, confirming the validity of the derived relationships. The selected final design ($N_s = 30$, $N_p = 15$, and $N_r = 60$) was used for motion validation under the same operating cases evaluated in the Python model.

5 Manufacturing and Assembly

All components of the planetary gear mechanism were fabricated using a fused deposition modeling (FDM) 3D printer. PLA-CF filament was selected due to its improved stiffness, dimensional stability, and suitability for rapid prototyping applications.

The parts were designed with appropriate clearances to ensure proper meshing and rotation of the gears after printing. Standard print settings were used, and care was taken to maintain dimensional accuracy in the gear teeth to minimize interference and excessive backlash.

Following fabrication, the components were assembled by placing the planet gears onto the carrier and aligning them with the sun and ring gears. The assembly was designed to be self-contained, allowing smooth rotation of all components without the need for additional fasteners or lubrication.

The final assembled mechanism demonstrated proper gear engagement and continuous motion, confirming that the manufacturing and assembly process was successful.

Following assembly, the mechanism was manually tested to verify its kinematic behavior. The input member was rotated while observing the resulting motion of the output components for each operating case. The observed directions of rotation and relative speed changes were consistent with the analytical and computational predictions, particularly the reduced carrier speed in the ring-fixed case and increased output speed in the sun-fixed case. Minor deviations from ideal behavior were present due to friction, gear backlash, and manufacturing tolerances inherent to the 3D printing process; however, the overall system performance closely matched the expected theoretical behavior.

6 Conclusion

This project successfully demonstrated the design, analysis, and fabrication of a 3D-printed planetary gear mechanism. The governing speed and torque relationships were derived analytically and implemented in a Python-based computational model to evaluate system behavior under multiple operating conditions.

The selected gear set effectively demonstrated speed reduction and torque amplification while satisfying geometric constraints required for proper meshing. Motion study results and physical testing confirmed that the behavior of the fabricated mechanism closely matched theoretical predictions.

Overall, this project highlights the effectiveness of combining analytical modeling, computational tools, and additive manufacturing to design and validate mechanical systems.

7 Appendix

7.1 Physical Pictures

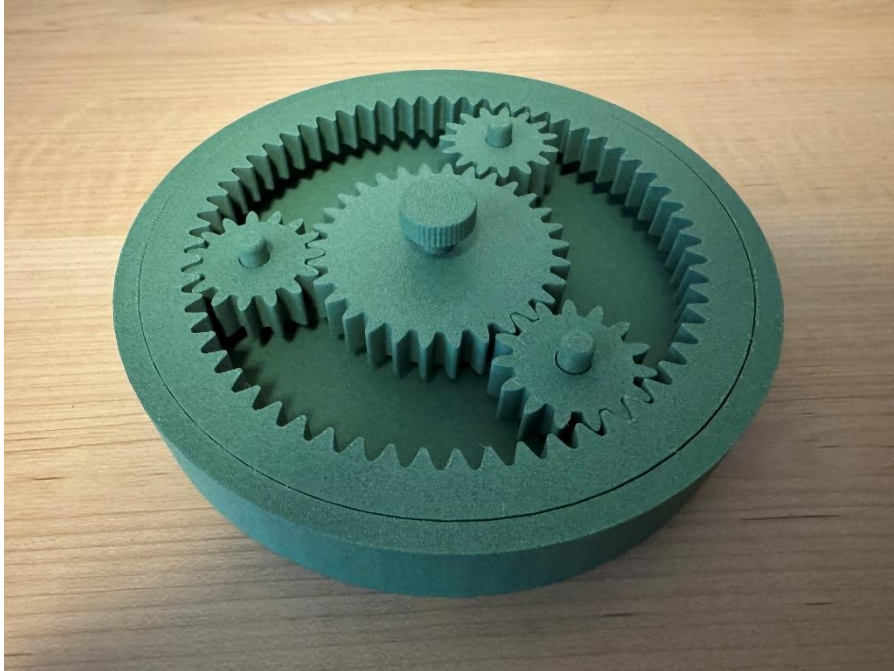


Figure 7.1.1: Physical View 1 Drawing



Figure 7.1.2: Physical View 2 CAD Drawing

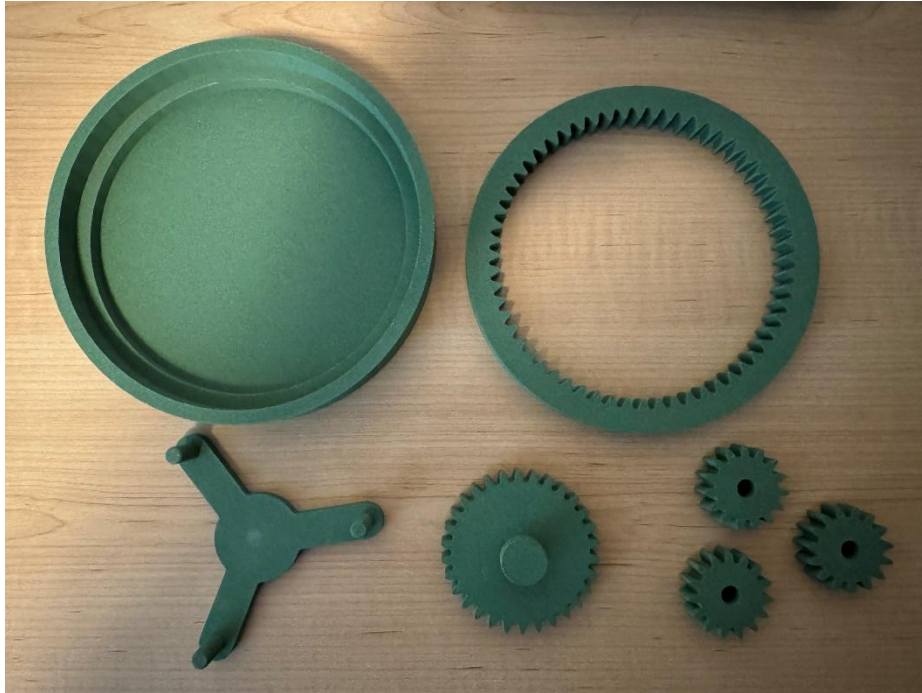
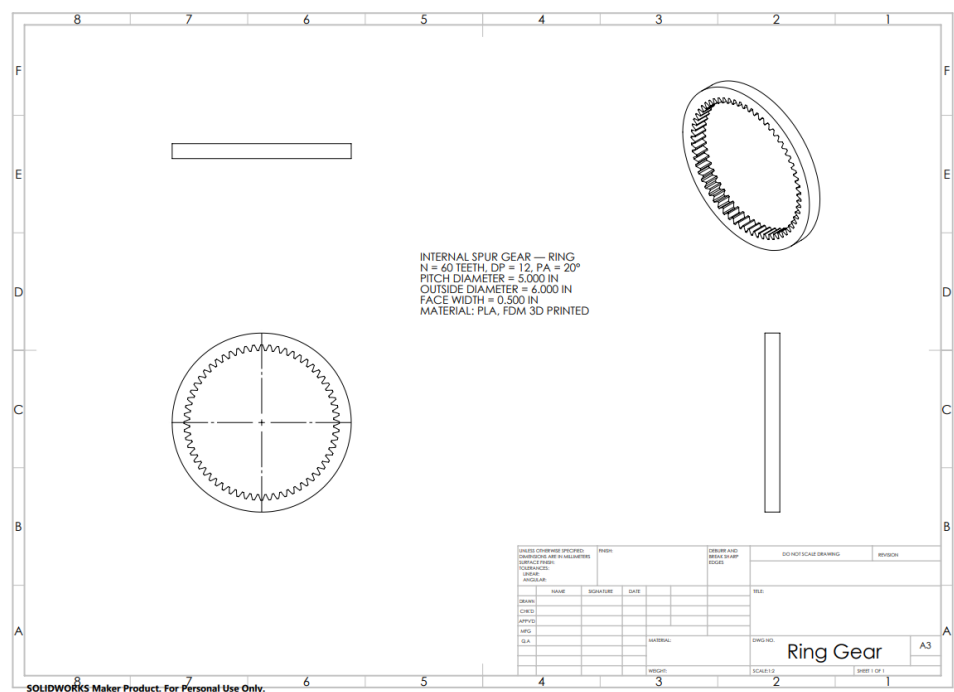
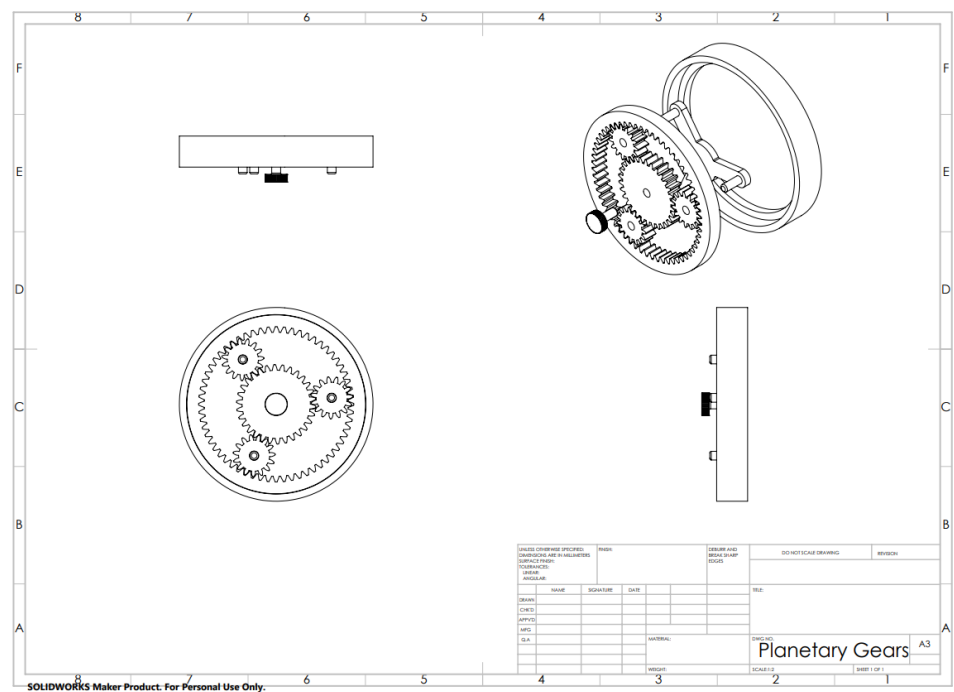
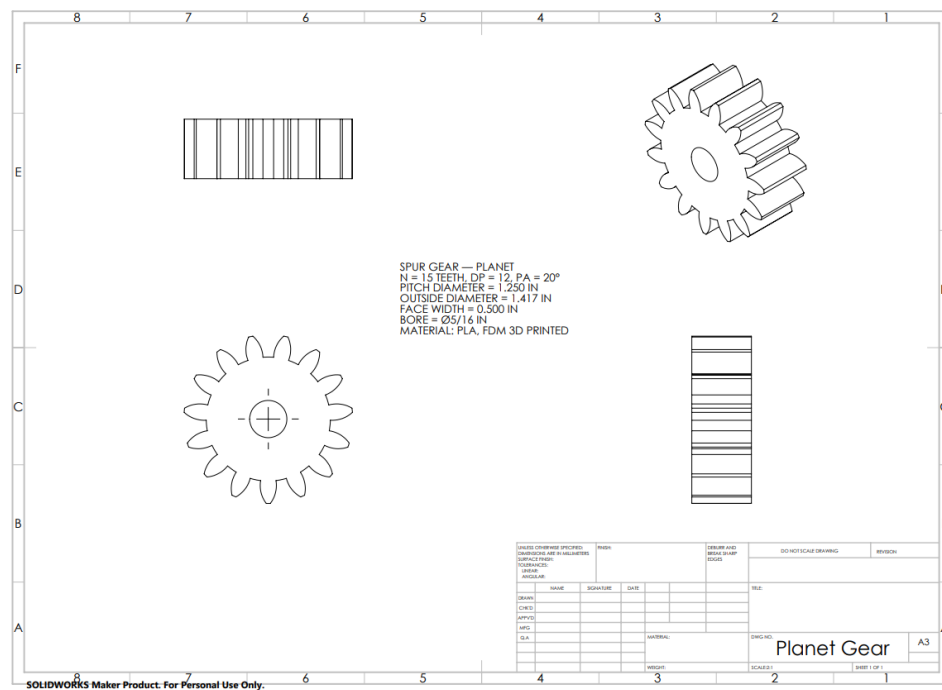
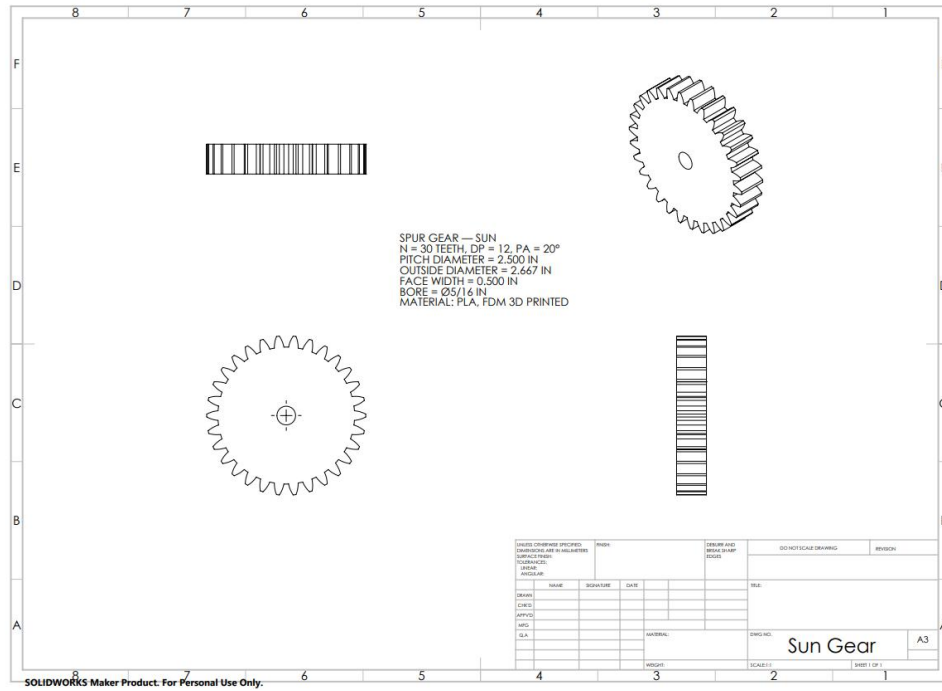


Figure 7.1.3: Parts View CAD Drawing

7.2 CAD Drawings





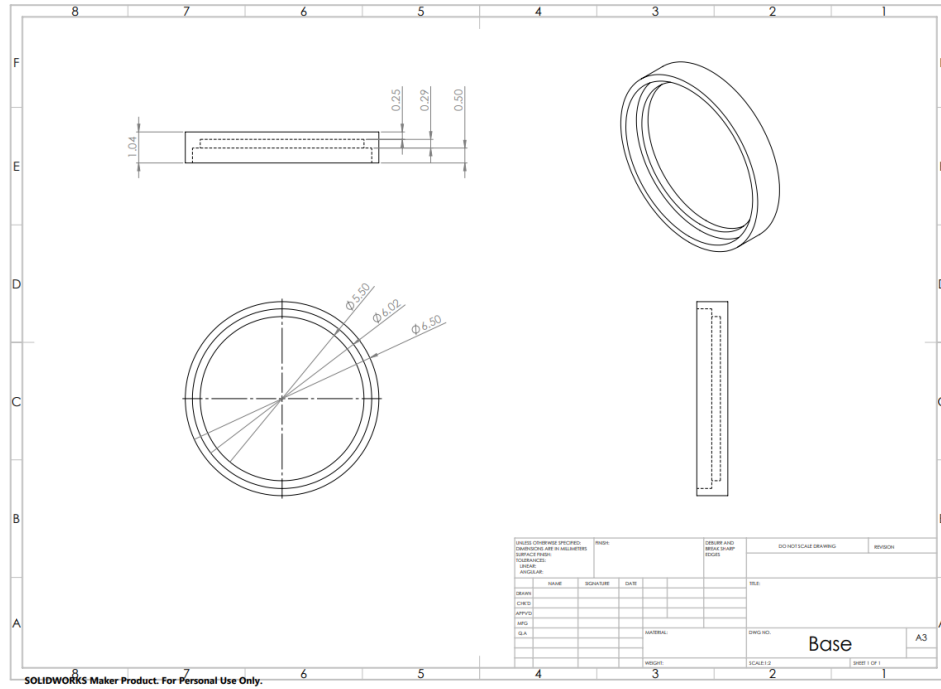


Figure 7.2.5: Base CAD drawing

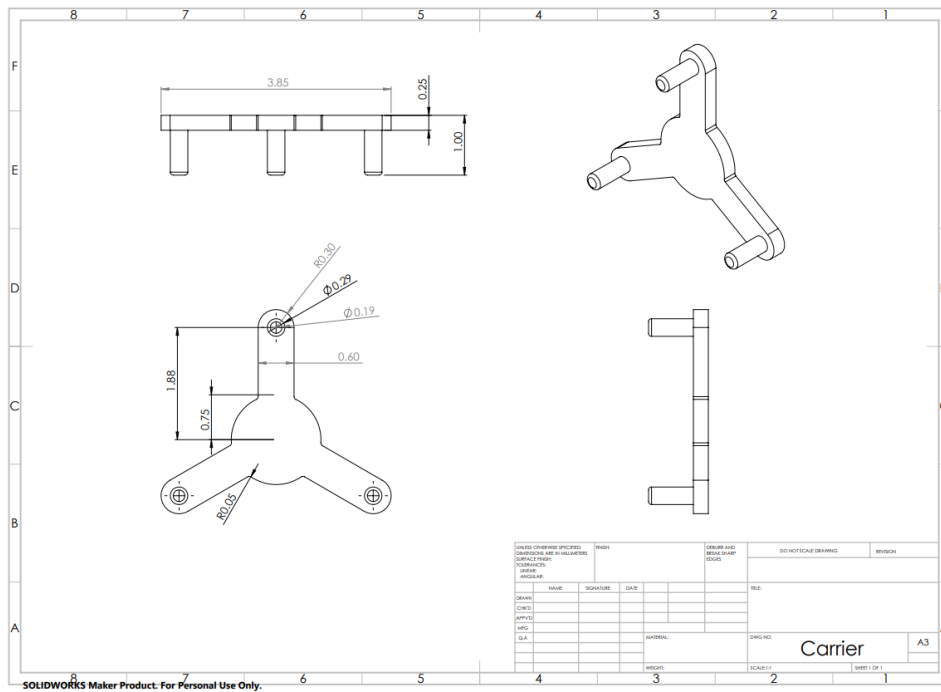


Figure 7.2.6: Carrier CAD drawing

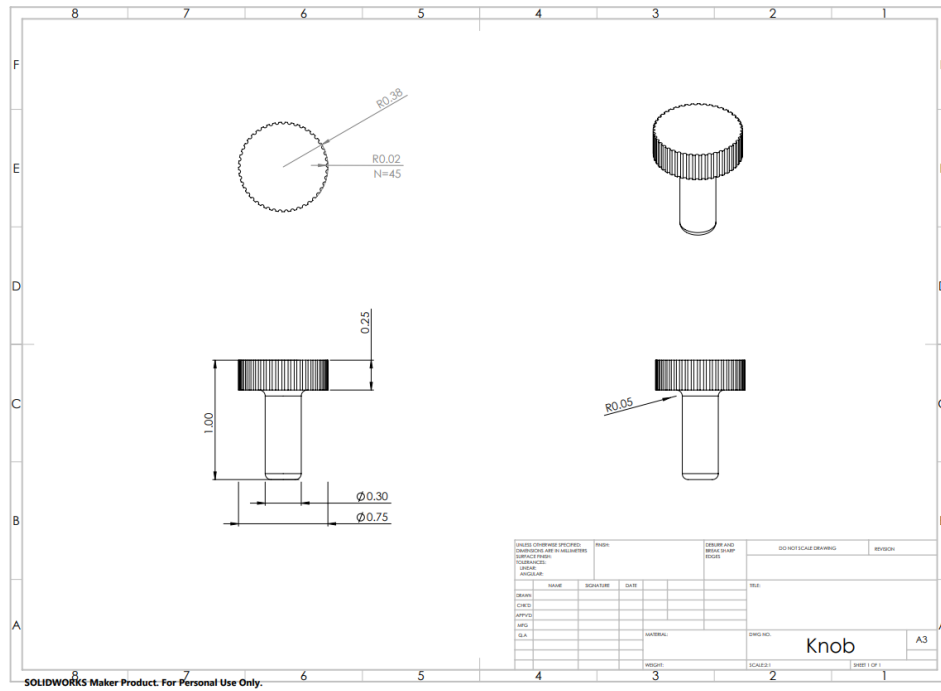


Figure 7.2.7: Knob CAD drawing

7.3 Python Code

```
# planetary_gear_analysis.py
# Computational model for a simple planetary gear mechanism
# Includes 3 standard operating cases + plots

import numpy as np
import matplotlib.pyplot as plt
from dataclasses import dataclass

@dataclass
class PlanetaryGearSet:
    Ns: float # sun teeth
    Np: float # planet teeth
    Nr: float # ring teeth

    def geometry_valid(self, tol=1e-9):
        return abs(self.Nr - (self.Ns + 2 * self.Np)) < tol

    def geometry_message(self):
        if self.geometry_valid():
            return f"Geometry is valid: Nr = Ns + 2*Np = {self.Nr}"
        return f"Geometry is NOT valid: expected Nr = {self.Ns + 2*self.Np}, got {self.Nr}"

    # Main planetary speed equation:
    #  $(\omega_s - \omega_c)/(\omega_r - \omega_c) = -N_r/N_s$ 

    def solve_wr(self, ws, wc):
        return wc - (self.Ns / self.Nr) * (ws - wc)

    def solve_ws(self, wr, wc):
        return wc - (self.Nr / self.Ns) * (wr - wc)

    def solve_wc(self, ws, wr):
        return (self.Ns * ws + self.Nr * wr) / (self.Ns + self.Nr)

    def torque_ratio_ring_to_sun(self):
        return self.Nr / self.Ns

def direction_str(omega):
    if omega > 0:
        return "ccw"
```

```

elif omega < 0:
    return "cw"
return "stationary"

def print_speed(label, omega):
    print(f"{label}: {abs(omega):.3f} rpm {direction_str(omega)}")

def print_case_header(title):
    print("\n" + "=" * 65)
    print(title)
    print("=" * 65)

# -----
# USER DESIGN INPUTS (EDIT THESE)
# -----
Ns = 30
Np = 15
Nr = 60

gearset = PlanetaryGearSet(Ns=Ns, Np=Np, Nr=Nr)

print_case_header("PLANETARY GEAR SET SUMMARY")
print(f"Sun gear teeth (Ns)   = {Ns}")
print(f"Planet gear teeth (Np)= {Np}")
print(f"Ring gear teeth (Nr)   = {Nr}")
print(gearset.geometry_message())
print(f"Relative speed ratio (ws-wc)/(wr-wc) = {-Nr/Ns:.3f}")
print(f"Ideal torque ratio Tr/Ts = {gearset.torque_ratio_ring_to_sun():.3f}")

# -----
# CASE 1: RING FIXED
# wr = 0, input = sun, output = carrier
# -----
print_case_header("CASE 1: RING FIXED (wr = 0), INPUT = SUN, OUTPUT = CARRIER")

ws_case1 = -100.0 # input sun speed (cw)
wr_case1 = 0.0    # ring fixed
wc_case1 = gearset.solve_wc(ws=ws_case1, wr=wr_case1)

print_speed("Sun speed (input)", ws_case1)
print_speed("Ring speed (fixed)", wr_case1)
print_speed("Carrier speed (output)", wc_case1)

```

```

# Gear reduction ratio
ratio_case1 = abs(ws_case1 / wc_case1) if wc_case1 != 0 else np.inf
print(f"Speed reduction ratio = {ratio_case1:.3f}")

# -----
# CASE 2: SUN FIXED
# ws = 0, input = carrier, output = ring
# -----
print_case_header("CASE 2: SUN FIXED (ws = 0), INPUT = CARRIER, OUTPUT = RING")

ws_case2 = 0.0
wc_case2 = -100.0 # input carrier speed (cw)
wr_case2 = gearset.solve_wr(ws=ws_case2, wc=wc_case2)

print_speed("Sun speed (fixed)", ws_case2)
print_speed("Carrier speed (input)", wc_case2)
print_speed("Ring speed (output)", wr_case2)

ratio_case2 = abs(wc_case2 / wr_case2) if wr_case2 != 0 else np.inf
print(f"Speed ratio = {ratio_case2:.3f}")

# -----
# CASE 3: CARRIER FIXED
# wc = 0, input = sun, output = ring
# -----
print_case_header("CASE 3: CARRIER FIXED (wc = 0), INPUT = SUN, OUTPUT = RING")

wc_case3 = 0.0
ws_case3 = -100.0 # input sun speed (cw)
wr_case3 = gearset.solve_wr(ws=ws_case3, wc=wc_case3)

print_speed("Carrier speed (fixed)", wc_case3)
print_speed("Sun speed (input)", ws_case3)
print_speed("Ring speed (output)", wr_case3)

ratio_case3 = abs(ws_case3 / wr_case3) if wr_case3 != 0 else np.inf
print(f"Speed ratio = {ratio_case3:.3f}")

# -----
# TORQUE EXAMPLE
# -----
print_case_header("TORQUE RELATIONSHIP EXAMPLE")

Ts = 10.0 # input torque on sun, N*m

```

```

Tr = gearset.torque_ratio_ring_to_sun() * Ts
Tc = -(Ts + Tr)

print(f"Input sun torque Ts = {Ts:.3f} N*m")
print(f"Ring torque Tr      = {Tr:.3f} N*m")
print(f"Carrier torque Tc   = {Tc:.3f} N*m")

# -----
# GRAPH 1: OUTPUT SPEED VS INPUT SPEED
# -----
input_speeds = np.linspace(-300, 300, 100)

# Case 1: wr = 0, solve wc from ws
case1_outputs = [gearset.solve_wc(ws=w, wr=0.0) for w in input_speeds]

# Case 2: ws = 0, solve wr from wc
case2_outputs = [gearset.solve_wr(ws=0.0, wc=w) for w in input_speeds]

# Case 3: wc = 0, solve wr from ws
case3_outputs = [gearset.solve_wr(ws=w, wc=0.0) for w in input_speeds]

plt.figure(figsize=(10, 6))
plt.plot(input_speeds, case1_outputs, label="Case 1: Ring Fixed (output = carrier)")
plt.plot(input_speeds, case2_outputs, label="Case 2: Sun Fixed (output = ring)")
plt.plot(input_speeds, case3_outputs, label="Case 3: Carrier Fixed (output = ring)")
plt.axhline(0, linewidth=1)
plt.axvline(0, linewidth=1)
plt.xlabel("Input Speed (rpm)")
plt.ylabel("Output Speed (rpm)")
plt.title("Planetary Gear Output Speed vs Input Speed")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

# -----
# GRAPH 2: OUTPUT TORQUE VS INPUT TORQUE
# -----
input_torques = np.linspace(0, 50, 100)
output_ring_torque = gearset.torque_ratio_ring_to_sun() * input_torques
output_carrier_torque = -(input_torques + output_ring_torque)

plt.figure(figsize=(10, 6))

```

```
plt.plot(input_torques, output_ring_torque, label="Ring Torque")
plt.plot(input_torques, output_carrier_torque, label="Carrier Torque")
plt.xlabel("Input Sun Torque (N*m)")
plt.ylabel("Output Torque (N*m)")
plt.title("Planetary Gear Torque Relationships")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```